

Master Thesis

High-Frequency Volatility Forecasting with an Option-Data-Augmented Neural Network

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Abstract

Accurate volatility forecasts are important for derivative pricing, portfolio allocation, and financial risk management. Although traditional volatility models primarily rely on historical returns, option prices contain forward-looking information about market expectations of future uncertainty. This thesis investigates whether high-frequency option-market information can improve short-horizon volatility forecasts when incorporated into a neural-network framework. The empirical analysis uses one-minute S&P 500 index data and option quotes from February to December 2025. Annualised realised volatility over five-minute intervals is forecast using a standard GARCH(1,1) model, a GARCH-X model with lagged at-the-money implied volatility, a Gated Recurrent Unit neural network using recent open-high-low-close observations, and an augmented neural network that additionally includes return-, calendar-, range-, and option-market features.

The out-of-sample results show that the GARCH-X model does not improve upon the standard GARCH model, despite the implied-volatility coefficient being statistically significant. Both neural-network specifications outperform the GARCH benchmark in the main test sample. The standard GARCH model obtains an MAE of 0.0357, an RMSE of 0.0471, and an R^2 of 0.4833. The OHLC-only neural network improves these values to 0.0275, 0.0426, and 0.5074, respectively, while the augmented neural network achieves the strongest performance with an MAE of 0.0271, an RMSE of 0.0411, and an R^2 of 0.5412. A Diebold–Mariano test rejects equal predictive accuracy between the two neural-network specifications indicating that the augmented model has significantly lower squared forecast loss. However, the reductions in MAE and RMSE are small, implying that the practical magnitude of the improvement is modest.

Robustness tests show that the neural network retains a moderate advantage after excluding the April 2025 volatility spike, but its performance is less robust across alternative evaluation settings. In particular, model rankings are sensitive to winsorization and to the separation between high- and low-volatility regimes, where the advantage of the neural network becomes less consistent. Overall, the findings indicate that neural networks can improve high-frequency volatility forecasts relative to traditional GARCH models. The engineered features provide additional predictive value in the examined test sample, although the improvement cannot be attributed exclusively to option-market information and may depend on the selected sample and prevailing market conditions.

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1 Introduction

1.1 Options

Options are a central segment of modern financial markets, the scale and liquidity of both exchange-traded and over-the-counter derivatives markets have made option prices attractive sources of information about market expectations and risk premia. For instance, the Options Clearing Corporation reports that it cleared 12,224,227,606 listed options contracts in 2024, underscoring the depth of exchange-traded options activity (The Options Clearing Corporation, 2025). In parallel, the Bank for International Settlements estimates that the notional value of outstanding OTC derivatives reached approximately \$846 trillion at end-June 2025 (Bank for International Settlements, 2025). In the United States alone, average daily options trading volume reached approximately 59 million contracts in 2025, highlighting the scale and liquidity of modern options markets (Cboe Global Markets, 2025). Beyond their size, option markets are particularly important because option prices depend directly on expectations about the future uncertainty of the underlying asset. As a result, option prices embed information about market participants' beliefs regarding volatility.

1.2 Volatility

Volatility plays a fundamental role in asset pricing, portfolio allocation and risk management. It measures the uncertainty surrounding future asset price movements and therefore serves as a key indicator of financial market risk. As a definition: *Volatility is a statistical measurement of the degree of variability of the return of a security or market index* (Investopedia, 2024). The importance of volatility is particularly evident in derivative markets, where option prices depend directly on expectations about future volatility.

Volatility is not directly observable, it must be estimated from available market data. In practice, this can be done using information from the underlying asset or by extracting information from derivative prices such as options. Both approaches attempt to recover the latent volatility process that drives asset price dynamics (Hull, 2018).

1.3 Research Motivation and Question

Despite the informational richness of option markets, extracting reliable volatility forecasts from high-frequency option data remains challenging. High-frequency financial data are affected by market

microstructure noise, bid-ask spreads, asynchronous trading, and varying liquidity across contracts. These factors can distort observed prices and complicate the extraction of meaningful signals.

At the same time, most traditional volatility forecasting models, such as GARCH-type models, rely primarily on historical return data and may fail to fully incorporate the potential forward-looking information embedded in option prices. While the econometric literature has developed a wide range of volatility estimators based on high-frequency returns, comparatively fewer studies focus on integrating high-frequency option data into forecasting frameworks.

Recent advances in machine learning provide new opportunities to address this gap. Neural networks (NN), in particular, are well-suited to capturing complex, nonlinear relationships in high-dimensional data and may therefore be able to better exploit the information contained in option markets.

Motivated by these developments, this thesis investigates whether volatility forecasts can be improved by combining high-frequency option data with machine learning techniques. The empirical analysis focuses on the S&P 500 index (SPX), using both option quotes across different strikes and maturities and open-high-low-close (OHLC) data of the underlying index.

The central research question of this study is:

Can a neural network framework augmented with high-frequency option data improve volatility forecasts relative to traditional GARCH models?

To answer this question, the following sub-questions are addressed:

1. How well do GARCH and GARCH-X forecast volatility?
2. To what extent does including option-based features improve neural network forecasts relative to using only underlying asset OHLC/return data?
3. Does a neural network model that incorporates both option data and OHLC data outperform traditional econometric models in forecasting volatility?

1.4 Methodological Overview

To address the research question, this study constructs volatility forecasts using both traditional econometric models and a machine learning approach. First, the GARCH benchmark model is estimated using data of the SPX. Then the GARCH-X model is calibrated with the at-the-money

(ATM) Black and Scholes implied volatility (IV) as exogenous variable. These models serve as a benchmark for evaluating forecasting performance.

Next, an NN model is developed that combines recent OHLC observations of the underlying index with engineered features derived from high-frequency option quotes. The option-based features include at-the-money implied volatility for selected maturities, total option volume, and total open interest. Additional return-, calendar-, and range-based features are also included to provide broader information on recent market conditions and potentially improve forecast accuracy. Together, these variables provide forward-looking and historical information without requiring the model to process the complete cross-sectional option-price surface.

The performance of the NN model is evaluated against the benchmark models using standard forecast accuracy metrics. This comparison provides insight into whether the inclusion of option market information and the use of flexible machine learning techniques lead to improved volatility forecasts. At the same time, the comparison highlights an important methodological trade-off. GARCH-type models can be regarded as relatively interpretable ‘open-box’ models, as their structure and parameters have clear economic and statistical meanings. In contrast, the NN may capture more complex and nonlinear relationships, but its internal decision-making process is less transparent. Consequently, the models are evaluated not only in terms of predictive accuracy, but also in light of the trade-off between forecast performance and interpretability.

Finally, the stability of the empirical findings is examined through a set of robustness tests. These tests assess whether the relative forecasting performance of the models remains consistent after excluding the April 2025 volatility spike, applying winsorised evaluation metrics, and separating the test sample into high- and low-volatility regimes. By evaluating model performance under alternative sample definitions and market conditions, the robustness analysis provides insight into whether the observed forecasting gains of the NN are stable or partly driven by specific observations and periods of market stress.

1.5 Contribution

This thesis contributes to the literature on volatility forecasting and option markets in several ways. First, it explores the use of high-frequency option data as an input for volatility forecasting, rather than focusing solely on return-based measures. Second, it applies an NN framework to integrate information from both option markets and underlying asset prices, allowing for nonlinear interactions

between these data sources. Third, the study provides an empirical comparison between machine learning models and traditional econometric approaches, specifically GARCH models. Finally, it evaluates the incremental value of option-based information in improving forecast accuracy, thereby contributing to the understanding of how forward-looking market data can enhance volatility modelling.

1.6 Thesis Outline

The remainder of this thesis is structured as follows. Section 2 reviews the relevant literature and identifies the research gap. Section 3 describes the data, feature construction, forecasting models, and experimental setup. Section 4 presents the empirical results and model comparisons. Section 5 evaluates the robustness of the findings. Section 6 concludes by answering the research questions and discussing limitations and directions for future research.

2 Literature Review

2.1 *Stylised Facts of Financial Volatility*

Volatility is a central concept in financial markets and represents the uncertainty or variability of asset returns (Campbell, Lo, & MacKinlay, 1997). Cont (2001) shows that volatility exhibits several stylised facts. First, volatility is time-varying and tends to cluster, meaning that periods of high volatility are followed by high volatility and vice versa. Second, volatility is not directly observable and must be inferred from return data or option prices. Third, volatility often exhibits long memory, meaning that shocks can persist over long horizons.

In addition, Cont (2001) highlights that financial return distributions deviate strongly from the normal distribution. Returns typically exhibit excess kurtosis, meaning they are more peaked around the mean and have significantly fatter tails, implying that extreme events occur more frequently than predicted by a Gaussian framework. Together, these stylised facts demonstrate that simple assumptions such as constant volatility and normally distributed returns are insufficient.

For the present thesis, these stylised facts are important for two reasons. First, they justify benchmark models such as GARCH, which are specifically designed to capture time-varying conditional variance. Second, they motivate the use of option-based features and flexible nonlinear models, since the information set relevant for future volatility is likely richer and more complex than what can be captured by linear models based solely on lagged returns.

2.2 *Volatility Estimators*

Because volatility is latent and thus cannot be directly observed, empirical research relies on estimators constructed from observed prices. The simplest estimator is the close-to-close estimator, which computes the standard deviation of log returns over a rolling window. Although easy to implement, this measure uses only closing prices and ignores information generated during the trading day.

To improve efficiency, range-based estimators such as Parkinson (1980), Garman and Klass (1980), and Rogers and Satchell (1991) incorporate high, low, open, and close prices. These estimators use more information from within the trading period and are generally more efficient than the close-to-close measure under specific assumptions.

A further refinement is the Yang–Zhang estimator (Yang & Zhang, 2000), which combines overnight return variance, intraday volatility, and range-based measures. It is designed to account

for opening-price jumps while remaining independent of price drift. Yang and Zhang (2000) show that it has the smallest variance among the considered estimators with comparable properties.

Despite these improvements, all of the above estimators rely on daily data and therefore fail to capture the rich intraday dynamics present in high-frequency financial markets. Given the high-frequency nature of the data used in this study, volatility is measured using realised volatility constructed from intraday returns. This approach allows for a more accurate representation of intraday price variation.

The annualised realised volatility is computed as:

$$r_{t,i} = \ln \left(\frac{P_{t,i}}{P_{t,i-1}} \right) \quad (1)$$

$$RV_t = \sum_{i=1}^N (r_{t,i})^2 \quad (2)$$

$$\sigma_t = \sqrt{RV_t \cdot 252 \cdot 78} \quad (3)$$

where P denotes the asset price at intraday time i , and N is the number of intraday observations within day t . The summation term represents the realised variance, obtained by aggregating squared high-frequency returns (e.g., 1-minute or 5-minute returns). Taking the square root converts variance into volatility, and multiplying by 252 annualises the measure under the assumption of 252 trading days per year. For 5-minute returns, the number of intraday observations is 78, corresponding to the number of 5-minute intervals in a standard U.S. trading day of 6.5 hours.

Importantly, the realised variance is constructed without subtracting the mean return. This differs from traditional variance estimators but is justified in high-frequency settings. As shown by Andersen and Bollerslev (1998), intraday returns have negligible mean relative to their variance, such that the contribution of the mean to total variation is effectively zero. Similarly, Barndorff-Nielsen and Shephard (2002) demonstrate that the sum of squared high-frequency returns provides a consistent estimator of the integrated variance of a continuous-time price process, without requiring mean adjustment. As the sampling frequency increases, the drift component becomes asymptotically irrelevant, and realised variance converges to the quadratic variation of the underlying process. The resulting measure is used as the target variable throughout the empirical analysis.

2.3 Option Prices and Implied Volatility

Option prices embed information about market uncertainty that is not available in historical return data alone. Under standard no-arbitrage assumptions, each observed option price can be mapped to an IV by inverting the Black-Scholes pricing formula (Black & Scholes, 1973). Specifically, for an observed market price C^{mkt} of a European call with strike K and maturity T , the implied volatility σ^{IV} is the unique value of σ satisfying

$$C^{\text{mkt}} = C_{\text{BS}}(S_0, K, T; \sigma^{\text{IV}}), \quad (4)$$

where C_{BS} denotes the Black-Scholes call price. Because this equation has no closed-form solution for σ , IV is computed numerically via root-finding methods such as the Newton-Raphson algorithm.

More generally, the volatility of the underlying asset need not be constant. Under the risk-neutral measure, the asset-price process may be written as

$$dS_t = (r_t - q_t)S_t dt + \sigma_t S_t dW_t^{\mathbb{Q}}, \quad (5)$$

where r_t is the risk-free rate, q_t is the continuous dividend yield, $W_t^{\mathbb{Q}}$ is a Brownian motion under the risk-neutral measure, and σ_t is an adapted stochastic volatility process. A standard regularity condition is that $\int_0^T \sigma_t^2 dt < \infty$ almost surely, ensuring that the accumulated variance over the option's life is finite. In this setting, the IV obtained from the Black-Scholes formula should not be interpreted as the actual instantaneous volatility at time t . Rather, it is the constant volatility input that reproduces the observed market price and therefore summarises, in a single number, the market's risk-neutral assessment of the future path of σ_t , together with the compensation investors require for bearing volatility risk.

The practical value of IV lies in standardisation: every option price, regardless of its strike or maturity, can be expressed as a single comparable volatility number. At-the-money (ATM) IV for a given expiry is widely interpreted as the market's consensus expectation of realised volatility over the remaining life of the option, adjusted for a variance risk premium (Christensen & Prabhala, 1998; Poon & Granger, 2003).

In practice, the Black-Scholes assumption of constant volatility is inconsistent with observed prices. Options at different strikes imply different values of σ^{IV} , producing the well-documented

volatility smile: a U-shaped pattern in which deep in-the-money and out-of-the-money options carry higher IV than ATM options (Figure 1). This smile reflects market pricing of tail risk and the leverage effect, and gave rise to stochastic-volatility models (Heston, 1993) and practitioner surface-fitting approaches such as SABR (Hagan, Kumar, Lesniewski, & Woodward, 2002). For the purposes of this thesis, however, these models are background: their key implication is simply that option prices encode information about expected future uncertainty in a way that historical-return models do not.

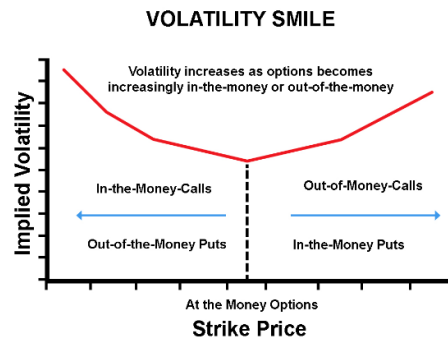


Figure 1: Volatility smile: implied volatility plotted against strike price for a fixed maturity.

The predictive content of IV for subsequent realised volatility is well established. Christensen and Prabhala (1998) show that ATM IV subsumes the information in historical volatility for forecasting realised volatility. Broader surveys confirm that option-implied measures frequently add information beyond time-series models based on returns alone (Kambouroudis, McMillan, & Tsakou, 2016; Poon & Granger, 2003). The empirical question in this thesis is therefore not whether IV matters, but whether its information — together with other option-market signals — can be exploited more effectively inside a nonlinear NN than in conventional econometric frameworks such as GARCH-X.

2.4 Classic Volatility Forecasting Models

2.4.1 GARCH – Conditional Volatility

Conditional volatility models estimate volatility based on past return dynamics. The most widely used model in this category is the GARCH(1, 1) model introduced by Bollerslev (1986). In this framework, returns are defined as

$$r_t = \mu + \varepsilon_t \quad (6)$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim (0, 1) \quad (7)$$

with conditional variance evolving according to

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8)$$

The model captures volatility clustering through persistence in the variance process. GARCH models have become a standard tool for forecasting volatility using historical return data.

Empirical evidence suggests that simple GARCH specifications often perform competitively relative to more complex volatility models (Poon & Granger, 2003). Although numerous extensions of the basic GARCH framework have been proposed to capture additional stylised facts of financial returns, their empirical forecasting improvements are often modest. Examples include the EGARCH model (Nelson, 1991), which captures asymmetric volatility effects; the GJR-GARCH model (Glosten, Jagannathan, & Runkle, 1993), which accounts for the leverage effect whereby negative returns increase volatility more than positive returns; and the IGARCH model (Engle & Bollerslev, 1986), which allows for highly persistent volatility dynamics.

A natural extension is to allow volatility to depend not only on past returns and past variance, but also on exogenous variables. This class of models is commonly referred to as GARCH-X. In the literature, the exogenous input can be macroeconomic variables, sentiment, realised measures, or option-based indicators such as IV. Hansen, Huang, and Shek (2012) Realised GARCH formulation explicitly describes a class of GARCH-X models with external measures, while later work formalises additive and multiplicative GARCH-X specifications.

2.4.2 HAR-RV Model

A prominent framework within volatility forecasting is the Heterogeneous Autoregressive model of Realised Volatility (HAR-RV), introduced by Corsi (2008). The model provides a reduced-form approach to modelling volatility dynamics using realised measures directly, motivated by the heterogeneous market hypothesis: market participants operating at different horizons generate persistence in volatility across multiple time scales. The standard specification models next-period realised volatility as a linear function of daily, weekly, and monthly components:

$$RV_{t+1} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1} \quad (9)$$

In contrast to GARCH-type models, where volatility is treated as a latent process, HAR-RV

directly models observed realised volatility. Although this makes it a natural benchmark for realised volatility forecasting, its daily, weekly, and monthly aggregation horizons do not map directly onto the five-minute setting examined here, and implementing it would require non-trivial redefinition of its core components. It is therefore not included as an empirical benchmark but represents a worthwhile direction for future research.

2.5 *Neural Networks in Finance*

The use of NNs in finance has expanded rapidly because financial data are nonlinear, noisy, and often high-dimensional (Ozbayoglu, Gudelek, & Sezer, 2020). Deep learning models can process large numbers of predictors and automatically learn nonlinear interactions that are hard to specify *ex ante* in standard econometric models. Reviews of deep learning in finance describe successful applications in asset pricing, return prediction, order book forecasting, volatility modelling, and risk prediction (Giantsidi & Tarantola, 2025).

A major argument for NNs is that they can capture complex cross-feature interactions. Gu, Kelly, and Xiu (2020) show that machine learning methods outperform traditional linear approaches largely because they can model nonlinearities and interactions among predictors. Although their focus is return prediction rather than volatility forecasting, the logic is directly relevant here: if option variables and OHLC variables interact in nonlinear ways, an NN may exploit that structure more effectively than GARCH or GARCH-X.

Recurrent neural networks (RNNs) remain a popular choice for sequence modelling. Among RNN architectures, recent evidence suggests that Gated Recurrent Units (GRUs) often provide a more efficient and equally effective alternative to Long Short-Term Memory (LSTM) networks. Hewamalage, Bergmeir, and Bandara (2021) show that GRUs typically match the predictive performance of LSTMs across a wide range of time-series forecasting tasks while requiring fewer parameters and exhibiting faster convergence. Given these characteristics, GRUs offer a strong balance between model complexity and forecasting accuracy, making them a suitable and computationally efficient choice for financial forecasting applications. Nguyen, Nguyen, Bui, Pham, and Nguyen (2026) concludes that GRU with the Adam optimiser algorithms consistently outperforms other models across all metrics.

A recent systematic review by Gunnarsson et al. (2024) concludes that artificial intelligence methods for predicting realised volatility and IV indices are promising, but that results vary with data construction, feature engineering, and evaluation design.

Overall, the deep-learning literature suggests a clear rationale for the empirical design of this thesis. NNs should be most useful when the feature set is large, heterogeneous, and potentially nonlinear. High-frequency OHLC data and high-frequency option market data fit that description closely. This makes a NN framework a natural candidate for testing whether option information can improve volatility forecasts beyond what traditional econometric benchmarks achieve.

2.6 Empirical Comparisons

2.6.1 GARCH and Neural-Network Forecasting Performance

Recent empirical research suggests that GARCH-type models remain important benchmarks in volatility forecasting, but that neural and hybrid methods often perform better when volatility dynamics are nonlinear, high-frequency, or structurally unstable. Rubio, Palacio Pinedo, Mejía Castaño, and Ramos (2023) study Tesla volatility using GARCH and related hybrid models. Their results show that GARCH remains a relevant econometric benchmark, with the GARCH specification reaching an R^2 of approximately 0.65 over a 21-day test horizon using one-step-ahead forecasts.

Evidence from neural-network-based studies generally points toward stronger predictive performance. Shi et al. (2022) forecast high-frequency volatility for 112 stocks and find that their model-fusion approach, based on an LSTM architecture, achieves an R^2 of approximately 0.86. Similarly, Mücher (2022) use an LSTM to transform intraday high-frequency returns into a daily volatility predictor. In an application to IBM stock, models combining this neural-network-based measure with past realised volatility outperform conventional HAR and MIDAS benchmarks. These findings suggest that LSTMs can extract nonlinear intraday patterns that are not fully captured by standard realised-volatility measures.

Further evidence is provided by Quang, Huyen, and Quyen (2025), who compare recurrent NNs, LSTMs, transformers, and ARMA-GARCH models for cryptocurrency volatility forecasting. In their Bitcoin application, the LSTM model reaches an R^2 of approximately 0.97, while the GARCH benchmark performs substantially worse. This suggests that recurrent neural architectures can be particularly effective in markets with strong nonlinear and short-term dynamics. However, this result should be interpreted cautiously, as the study appears to evaluate scaled closing-price dynamics rather than a pure volatility target, while the inclusion of the closing price as an input may make the prediction task substantially easier.

2.6.2 Architectural Choice

The forecasting performance of NNs depends strongly on the selected architecture. Recurrent models are particularly suitable for volatility forecasting because they can learn temporal dependencies from sequential financial data. Jain et al. (2026) compare LSTM and GRU models using S&P 500 and VIX data and report that the GRU model achieves an R^2 of 0.91, outperforming the LSTM specification in their setting. This result suggests that GRUs can capture relevant volatility dynamics while using a simpler architecture with fewer parameters than LSTMs.

Architectural flexibility is also important when combining multiple sources of information. Zhang, Zhang, Cucuringu, and Qian (2024) find that NNs trained on pooled intraday data outperform stock-specific models, particularly when market-wide volatility is included as an additional predictor. Their results indicate that neural architectures are well suited to learning common and nonlinear volatility patterns from richer input sets. Together, these findings support the use of a GRU-based framework that can model sequential dependencies while incorporating additional market features.

2.7 Research Gap

Despite extensive research on volatility modelling, an important gap remains in understanding how the information embedded in option markets can be effectively translated into volatility forecasts. Traditional econometric models may therefore be limited in their ability to fully exploit the high-dimensional structure of option market data.

The objective of this thesis is to address this gap by investigating whether NN models can use high-frequency option market observables to improve volatility forecasting. By comparing these approaches, the thesis evaluates the incremental forecasting value of high-frequency option data within a flexible machine learning framework.

3 Methodology

3.1 Data Collection

The option dataset used in this study consists of S&P 500 index (SPX) call and put option quotes over the period from February 3, 2025 to December 31, 2025. A data gap is present between March 17, 2025 and April 4, 2025 due to a temporary failure in the data collection process. This period is excluded from the analysis.

Consistent with standard market conventions, observations are only available for trading days. As such, weekends and U.S. market holidays are excluded. Specifically, no data are recorded on the following dates: Presidents' Day (February 17), Good Friday (April 18), Memorial Day (May 26), Juneteenth (June 19), Independence Day (July 4), Labor Day (September 1), and Thanksgiving Day (November 27).

Each option observation is recorded with a delay of approximately 22 seconds relative to the snapshot timestamp. This delay arises from the data provider's aggregation and dissemination process, where quotes are collected, processed, and then published with a short latency. While this is not expected to materially affect the analysis, it is important to acknowledge in a high-frequency setting, as it may introduce minor asynchrony between option prices and the underlying index. The raw option data structure is shown in Table 1.

Table 1: Option Data Structure and Example Values

Field	Example	Field	Example	Field	Example
Symbol	SPXW	Last	182.60	TradeSize	100
TradedMarket	31	TradeDate	2024-04-12	TradeTime	14:25:00
Open	180.10	High	183.20	Low	179.80
Close	182.55	Bid	182.45	BidMarket	31
BidSize	200	Ask	182.65	AskMarket	31
AskSize	180	Volume	3,450,000	PDayVolume	4,120,000
TradeCount	52,340	OpenInterest	12,450	ExpirationDate	2025-06-21
Strike	180	SnapshotTime	14:30:00		

Additionally, the underlying S&P 500 index data are available at a 1-minute frequency for the full calendar year 2025 within regular trading hours. This allows for consistent alignment between

option prices and the underlying asset by matching observations to the closest available timestamp. The data consist of an open, high, low, and close (OHLC) price series of the underlying index. The daily log returns are shown in Figure 2; additional data-exploration graphs can be found in Appendix B.

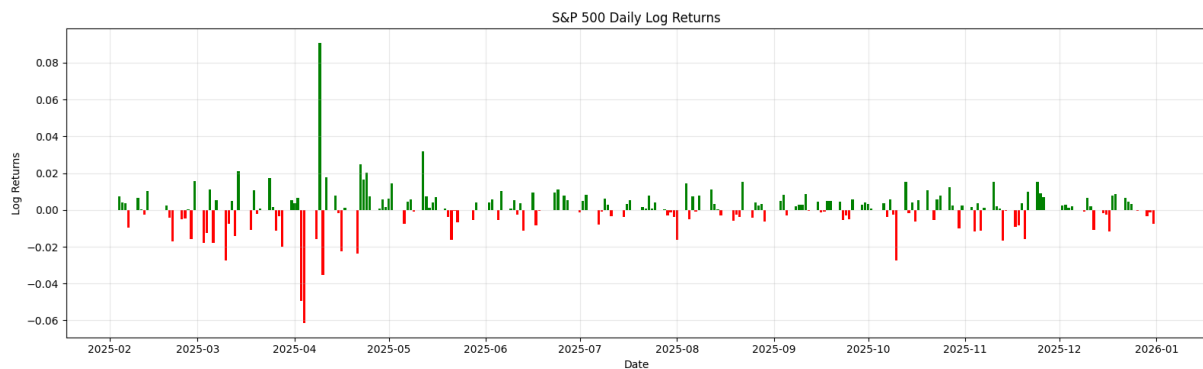


Figure 2: Daily Log Returns Derived from 1-Minute Data

The empirical analysis focuses on S&P 500 call and put options for several reasons. First, these contracts are among the most actively traded European-style options, providing high liquidity and reliable pricing. Second, the availability of daily dividend information for the index enhances the accuracy of option valuation. Finally, S&P 500 options have been extensively studied in the literature, facilitating comparability with prior research.

3.2 Feature Engineering

The data preprocessing pipeline was designed to transform two raw data sources into a clean time-series dataset suitable for volatility forecasting. The datasets were merged into a single chronologically ordered data frame, after which the forecasting features were constructed.

Several practical issues had to be addressed during preprocessing. First, the index and option data were not always perfectly synchronised, partly due to differences between European and U.S. timestamp conventions. Second, the transition between summer time and winter time created additional timestamp alignment problems. These issues were corrected to ensure that option quotes and index observations referred to the same market time. Third, particular attention was paid to preventing look-ahead bias. All predictive variables were lagged so that only information available before the forecasted interval could be used. Missing values created by lagged and rolling feature

construction were removed rather than imputed.

From the cleaned combined data frame, several lagged-return, calendar, range, and option-market features were constructed. These are shown in Table 2.

Table 2: Constructed Forecasting Features

Feature	Description
prev_5m_log_return_1	Previous 5-minute log return
prev_5m_log_return_2	Second previous 5-minute log return
prev_5m_log_return_3	Third previous 5-minute log return
prev_5m_log_return_4	Fourth previous 5-minute log return
prev_day_log_return	Previous trading-day log return
day_of_week	Weekday indicator
day_of_month	Day-of-month indicator
prev_5m_high_low_range	Previous 5-minute high-low range
prev_atm_iv_30dte	ATM IV for 30-day options
prev_atm_iv_0dte	ATM IV for 0-day options
prev_total_option_volume	Total option volume
prev_total_option_oi	Total option open interest
OHLC sequence	Previous five 1-minute OHLC observations

The selected features are chosen to capture both historical price dynamics and forward-looking option-market information. Lagged 5-minute returns and the previous high-low range summarise recent short-term movements in the underlying index, while the previous-day return captures broader daily market direction. Calendar variables account for possible day-specific patterns. The implied-volatility, volume, and open-interest variables incorporate information from the option market, reflecting market expectations, trading activity, and liquidity conditions. Finally, the OHLC sequence allows the NN to learn short-term temporal patterns directly from recent price evolution.

Several more complex feature specifications were also tested, including return transformations, volume-based variables, bid-ask spread measures, and option-surface characteristics such as implied-volatility slopes, skew, and smile curvature. However, these richer specifications did not improve forecasting performance consistently. Therefore, the final feature set was kept more parsimonious to reduce noise and avoid unnecessary model complexity.

For the OHLC NN, the OHLC values of the previous five 5-minute intervals were used as input. Since each interval contains four OHLC variables, this results in 20 input features. In the augmented NN, these 20 OHLC features were combined with the 12 engineered features, resulting in a total of 32 input features.

For the standard GARCH model, only the 5-minute log-return series is required. The GARCH-X model extends this specification by including an exogenous variable in the variance equation. Several alternatives were tested as exogenous inputs, including the low-high price range, trading volume, and implied volatility. However, the additional market-based variables did not lead to a consistent or significant improvement in forecasting performance. Therefore, the final GARCH-X specification uses only implied volatility as the exogenous input, as this provided the most parsimonious extension while preserving the direct link to market expectations of future volatility.

3.3 Neural Network Setup

Following the motivation for using GRU-based models discussed earlier (2.5), the NN is implemented as a stacked GRU architecture for one-step-ahead volatility forecasting. The model consists of two GRU layers, where the first layer returns the full sequence of hidden states and the second layer summarises this information into a final hidden representation. This representation is then passed through a fully connected dense layer with a ReLU activation function, followed by a linear output layer that produces the volatility forecast, as shown in Figure 3.

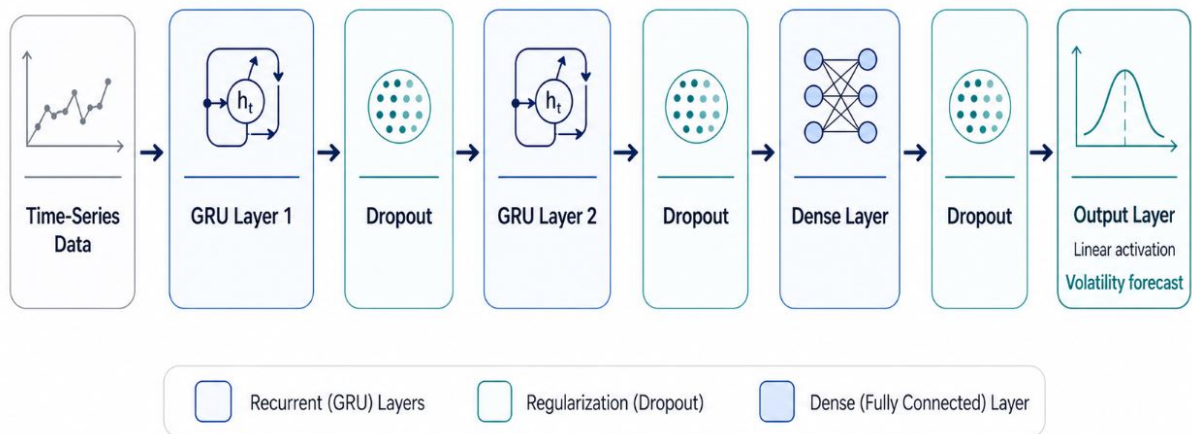


Figure 3: GRU-Based NN Architecture

Dropout is applied after both GRU layers and after the dense layer to reduce overfitting. The model is trained using the Adam optimiser, with mean absolute error (MAE) as the loss function. The final architecture was selected through a combination of empirical experimentation and guidance from the literature on NNs for financial time-series forecasting. Alternative specifications, including Huber and RMSE-based loss functions, were tested but did not improve overall performance. Therefore, the final model reflects both literature-based design choices and data-driven experimentation (Jain et al., 2026; Misra, Si, Wen, Zhou, & Han, 2025; Yachou, Abahman, & Hakimi, 2026).

The augmented model is implemented using a two-input architecture. The OHLC data are provided as a sequential input with shape $((\text{samples}, 5, 4))$, where the five time steps represent the previous five intervals and the four variables correspond to the OHLC prices. The 12 engineered features are provided through a separate input layer with shape $((\text{samples}, 12))$. This setup allows the GRU component to learn short-term temporal patterns from the OHLC sequence, while the engineered features are added separately as contextual information before making the final prediction. This means that the model has an additional dense layer.

Hyperparameter tuning was performed using Bayesian optimisation. The search procedure optimised the validation MAE over a limited number of candidate architectures. The tuned hyperparameters were the number of units in the two GRU layers, the number of neurons in the dense layer, the dropout rate, and the learning rate of the Adam optimiser. The search space and setup are shown in Table 3.

Hyperparameter Search Space	
Hyperparameter	Candidate Values
First GRU layer units	32, 64, 128
Second GRU layer units	32, 64, 128
Dense layer units	16, 32, 64
Dropout rate	0.0, 0.1, 0.2, 0.3
Learning rate	0.0001, 0.0005, 0.001
Fixed Training Setup	
Setting	Value
Batch size	512
Epochs per trial	200
Validation split	20%
Optimisation method	Bayesian optimisation
Objective	Validation MAE
Maximum number of trials	10

Table 3: Hyperparameter Search Space and Training Setup for the GRU Model

For the augmented model, the additional dense layer introduced after concatenating the OHLC sequence and engineered features is also included in the hyperparameter-tuning process.

Two additional methodological choices should be noted. First, the models are trained for 1000 epochs to give the network sufficient opportunity to learn the temporal patterns in the five-minute volatility data. The model updates its weights gradually, making a larger number of epochs appropriate. The training curves show that the loss decreases over time and eventually stabilizes, indicating that the chosen training length is sufficient for the model to reach a stable performance level.

Second, the final model is not necessarily taken from the last epoch. Instead, a model checkpoint callback is used during training, which monitors the validation loss after each epoch and saves the model weights only when a new lowest validation loss is achieved. Therefore, the reported performance is based on the model weights from the epoch with the best validation performance. This reduces the risk of selecting weights from a later epoch in which the model may have started to generalize less effectively to unseen data.

3.4 Implementation of GARCH and GARCH-X

The standard GARCH model is implemented using the `arch` Python package (Sheppard, 2021). A parsimonious GARCH(1,1) specification is used as the benchmark model, with normally distributed innovations. The Gaussian specification is chosen to keep the benchmark simple and directly comparable to the manually estimated GARCH-X model, which is also estimated using a Gaussian likelihood.

Returns are scaled before estimation to improve numerical stability and rescaled after forecasting. The model is estimated on the training sample only, after which fixed-parameter one-step-ahead forecasts are generated for the test period. Forecasted conditional variances are converted into volatility by taking the square root and are annualised using the same convention as the realised-volatility target.

The GARCH-X model extends this benchmark by adding lagged option IV to the variance equation:

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \beta h_{t-1} + \gamma x_{t-1}^2, \quad (10)$$

where x_{t-1} denotes the lagged IV input. The lag ensures that only information available before the forecast interval is used. Unlike the standard GARCH model, GARCH-X is estimated manually by minimising a Gaussian negative log-likelihood. The estimated parameter vector is:

$$\theta = (\mu, \omega, \alpha, \beta, \gamma). \quad (11)$$

During estimation, the restrictions $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\gamma \geq 0$, and $\alpha + \beta < 0.999$ are imposed. These constraints ensure positive conditional variance and prevent near-nonstationary GARCH dynamics. The conditional variance is initialised using the sample variance of the training returns, and a small lower bound is imposed during recursion to avoid numerical problems.

The main modelling choice is therefore the comparison between a return-only volatility benchmark and a specification that additionally incorporates option-market information. Since both models use the same forecast horizon, annualisation procedure, train–test split, and evaluation metrics, differences in performance can be attributed to the inclusion of the implied-volatility term rather than to differences in implementation.

3.5 Experimental Setup

The experimental setup follows a time-series forecasting framework with a strict chronological separation between training and test data. This chronological split was implemented explicitly through a fixed cut-off, where all observations up to Friday, 3 October 2025 were assigned to the training set and observations from Monday, 6 October 2025 onward were assigned to the test set. This setup ensures that the estimated models only use past information when generating out-of-sample forecasts.

Dataset	Start Date	End Date
Training set	2025-02-03	2025-10-03
Test set	2025-10-06	2025-12-31

Table 4: Training and Test Date Ranges

Model performance is evaluated by comparing forecasted volatility against realised volatility in both the training and test periods. Across the models, the same regression-style performance metrics are reported: MAE, root mean squared error (RMSE), and R^2 . MAE and RMSE measure forecast accuracy in absolute- and squared-error terms. R^2 summarises the proportion of variation in realised volatility explained by the forecasts. In addition, Diebold–Mariano tests are used to assess whether differences in out-of-sample forecast accuracy between competing models are statistically significant.

Additional robustness tests were performed to verify the out-of-sample forecasting ability of the models.

4 Results

4.1 Baseline

Before evaluating the GARCH and neural network models, two simple baseline forecasts are included. The first is a mean baseline, which predicts realised volatility using the average volatility observed in the training sample. The second is a last-value baseline, which uses the most recent observed realised volatility as the forecast for the next period. These results provide a reference point for assessing whether the more advanced models add predictive value beyond simple forecasting rules.

The mean baseline performs poorly, with an MAE of 0.0484, RMSE of 0.0624, and a negative R^2 of -0.0550. This indicates that using a constant average forecast is less accurate than simply predicting the mean of the test sample. The last-value baseline performs substantially better, with an MAE of 0.0331, RMSE of 0.0494, and an R^2 of 0.3399. This suggests that realised volatility is persistent over short intervals, making the most recent observation a relatively strong baseline for five-minute volatility forecasting.

4.2 GARCH Performance

4.2.1 Parameter Estimation

Table 5 presents the estimated parameters of the GARCH(1,1) model.

Parameter	Estimate	t-statistic	p-value
μ	9.8155×10^{-4}	0.893	0.372
ω	3.2151×10^{-4}	0.705	0.481
α_1	0.2002	1.740	0.0818
β_1	0.7798	5.281	1.282×10^{-7}

Table 5: GARCH(1,1) Parameter Estimates

The mean parameter μ is small and statistically insignificant, indicating that the average 5-minute return is not significantly different from zero. The constant variance term ω is also insignificant. The ARCH coefficient α_1 is positive and weakly significant at the 10% level, suggesting some short-term reaction of volatility to recent shocks. The GARCH coefficient β_1 is highly significant, indicating strong volatility persistence. Since $\alpha_1 + \beta_1 \approx 0.98$, volatility shocks decay slowly over time.

Table 6 presents the estimated parameters of the GARCH-X(1,1) model.

The mean parameter and constant variance term are again small and statistically insignificant.

Parameter	Estimate	t-statistic	p-value
μ	8.6712×10^{-6}	1.210	0.226
ω	1.0000×10^{-12}	0.000	1.000
α_1	0.0500	9.771	< 0.001
β_1	0.8407	75.013	< 0.001
γ	4.1349×10^{-7}	2.030	0.042

Table 6: GARCH-X(1,1) Parameter Estimates

The ARCH coefficient α_1 and GARCH coefficient β_1 are both positive and highly significant, showing that volatility responds to recent shocks and is strongly persistent over time. The exogenous implied-volatility coefficient γ is positive and statistically significant at the 5% level, suggesting that implied volatility contains additional information for explaining conditional variance. It must be noted that the variable is relatively small because the input IV is of a different order of magnitude than the log return series.

4.2.2 Forecasting results

To generate out-of-sample forecasts, one-step-ahead forecasts of conditional variance are produced for the entire test period and subsequently transformed into volatility forecasts. These forecasts are then compared against realised volatility to evaluate predictive performance. The out-of-sample performance of the model is summarised in Table 7.

Sample	Model	MAE	RMSE	R^2
Train	GARCH	0.0483	0.1034	0.4590
Train	GARCH-X	0.0487	0.0914	0.4698
Test	GARCH	0.0357	0.0471	0.4833
Test	GARCH-X	0.0412	0.0520	0.4703

Table 7: Forecasting Performance of GARCH and GARCH-X Models

This shows that the GARCH-X model performs slightly better than the standard GARCH model in-sample, with a lower RMSE of 0.0914 compared to 0.1034 and a marginally higher R^2 of 0.4698 compared to 0.4590. However, this improvement does not carry over to the test sample. Out-of-sample, the standard GARCH model achieves a lower MAE and RMSE than the GARCH-X model, indicating better forecasting accuracy on unseen data. Although the GARCH-X model has a slightly lower test R^2 , the difference is small. Overall, the results suggest that while the inclusion of the exogenous variable improves the model fit during training, it does not improve out-of-sample forecasting performance.

Given the absence of clear predictive gains and the additional complexity introduced by the exogenous variable, the simpler GARCH(1,1) specification is used in the remainder of the analysis.

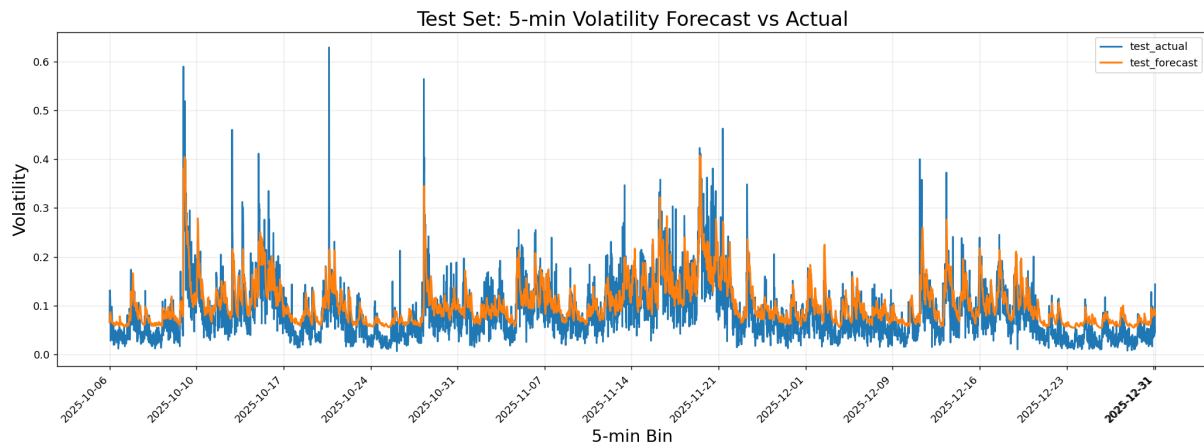


Figure 4: Realised Volatility vs GARCH(1,1) predicted

Figure 4 illustrates the comparison between the realised 5-minute volatility and the GARCH(1,1) forecast over the full test period. Periods of elevated volatility are clearly reflected in the forecast, indicating that the model successfully reproduces volatility clustering. However, the figure also highlights an important limitation: the GARCH model tends to produce smoother dynamics compared to realised volatility. In particular, sharp spikes in RV are often underestimated or lagged in the forecast. The figure also shows that forecasts are generally higher estimated in very low volatility areas. Additional error measures can be found in appendix C.

4.3 NN Performance

4.3.1 NN OHLC

The selected hyperparameter configuration consisted of 64 units in the first GRU layer, 64 units in the second GRU layer, 32 neurons in the dense layer, a dropout rate of 0.2, and a learning rate of 0.0005. The resulting model contained a total of 121,541 parameters.

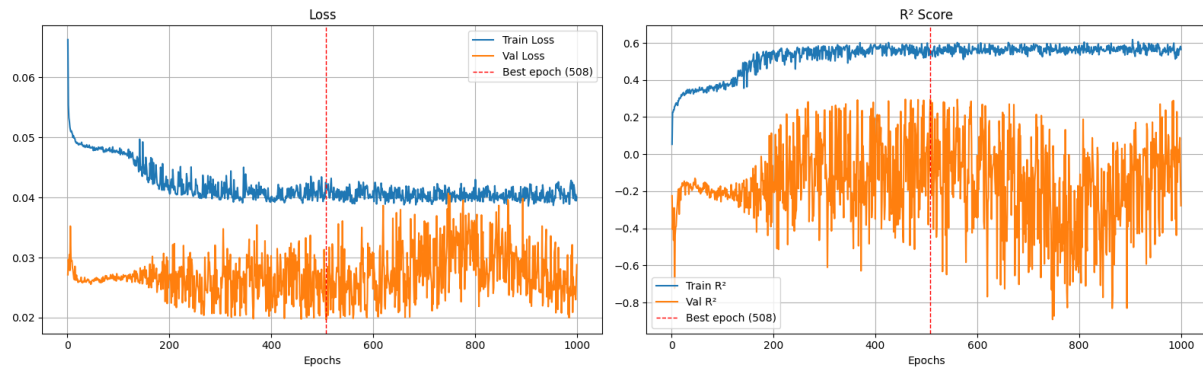


Figure 5: Train and Validation loss (MAE) and R^2 over the epochs NN OHLC

Figure 5 shows the training loss decreases steadily and the training R^2 increases to approximately 0.6, indicating that the model learns the training data well. However, the validation R^2 is highly volatile and often negative, suggesting weak out-of-sample performance. The validation loss reaches its minimum around epoch 508, which is therefore selected as the best epoch. The model achieves a loss and MAE of 0.0275, an RMSE of 0.0426, and an R^2 of 0.5074.

The volatile movements in the validation loss and validation R^2 can be explained by the nature of high-frequency volatility data. Since the model is trained on five-minute observations, small changes in market conditions can lead to relatively large changes in forecasting performance across validation batches. Volatility is also highly clustered and affected by sudden market movements, which makes the validation set less stable than the training set. In addition, the validation sample may contain periods with different volatility regimes than the training sample, causing the model to perform better in some epochs and worse in others.

The validation loss remains consistently below the training loss and does not cross above it, which suggests that there is no clear sign of overfitting. This may partly be explained by the use of dropout regularisation, which makes the training task more difficult during training, while dropout is disabled during validation. As a result, the model can achieve lower validation loss than training loss even when generalisation remains acceptable.

4.3.2 NN OHLC + Features

The selected specification consists of two GRU layers with 64 units, followed by a dense layer with 32 neurons. After concatenation with the engineered features, an additional dense layer with 64

neurons is used before the linear output layer. The dropout rate is 0.2 and learning rate 0.001. Total trainable parameters for this model are 130,277.

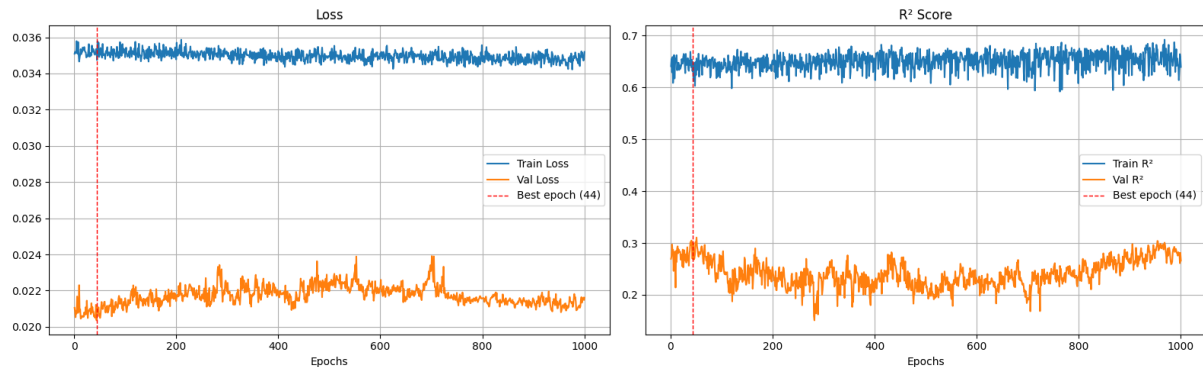


Figure 6: Train and Validation loss (MAE) and R^2 over the epochs NN + Features

Figure 6 shows the training loss remains stable around 0.035, while the validation loss stays lower at approximately 0.021–0.023. The training R^2 remains around 0.65, whereas the validation R^2 is positive but lower, fluctuating mostly between 0.20 and 0.30. The best epoch is reached early: epoch 44. Overall, the model generalizes better than the previous specification, although a clear gap between training and validation performance remains.

4.4 Overview of results

The out-of-sample results (see Table 8) show that both NN specifications outperform the GARCH benchmark across all three evaluation metrics. The GARCH model obtains an MAE of 0.0357, an RMSE of 0.0471, and an R^2 of 0.4833, while the OHLC NN improves these values to 0.0275, 0.0426, and 0.5074, respectively. The augmented neural network with additional engineered features achieves the strongest performance in this specific test set, with an MAE of 0.0271, an RMSE of 0.0411, and an R^2 of 0.5412. Since the target variable is annualised volatility, these errors can be interpreted as deviations of roughly 2.7 to 3.6 percentage points in annualised volatility terms. This suggests that the forecasts are reasonably close in magnitude, although the remaining RMSE indicates that meaningful forecast errors remain, which is realistic given the noisy and time-varying nature of volatility.

Table 8: Out-of-Sample Forecasting Performance Across Models

Model	MAE	RMSE	R ²
GARCH	0.0357	0.0471	0.4833
NN OHLC	0.0275	0.0426	0.5074
NN OHLC + Features	0.0271	0.0411	0.5412
Last Value Baseline	0.0331	0.0494	0.3399

To assess whether the differences in forecast accuracy are statistically significant, Diebold-Mariano tests are performed using out-of-sample squared forecast errors (Diebold & Mariano, 1995). First, the OHLC-only NN is compared with the augmented NN, with the loss differential defined as the squared error of the OHLC-only model minus that of the augmented model. The test produces a Diebold-Mariano statistic of 3.358 and a p-value of 0.0008. The positive mean loss differential of 0.000125 indicates that the OHLC-only model has a higher average squared forecast error. The null hypothesis of equal predictive accuracy is therefore rejected at the 1 percent significance level, providing evidence that the augmented NN forecasts volatility more accurately than the OHLC-only specification under squared-error loss.

Although the augmented neural network performs statistically better than the OHLC-only model, the reductions in MAE and RMSE are small, indicating limited practical significance. Moreover, the results may depend on the specific train-test split and prevailing market conditions. The findings should therefore be interpreted as evidence of a modest improvement rather than proof that the engineered features consistently produce meaningfully better forecasts.

Second, the augmented NN is compared with the standard GARCH(1,1) benchmark, with the loss differential defined as the squared error of the augmented NN minus that of GARCH. The resulting Diebold-Mariano statistic is -6.82 , with a p-value of 9.31×10^{-12} . The negative mean loss differential of -0.000397 indicates that the augmented NN produces systematically lower squared forecast errors than GARCH. The null hypothesis of equal predictive accuracy is therefore rejected at all conventional significance levels, providing strong statistical support for the conclusion that the GRU-based augmented NN outperforms the traditional GARCH benchmark in the examined test sample.

The Diebold-Mariano test assumes covariance-stationary forecast errors and non-nested models. While the NN and GARCH satisfy the non-nested condition, serial correlation in high-frequency

forecast errors may affect the test. The findings should therefore be interpreted as strong evidence rather than definitive proof of superiority.

5 Robustness Tests

To assess the robustness of the volatility forecasting results, several additional tests are conducted. First, the April 2025 tariff-related volatility spike is excluded from the training data. Second, the forecasts are evaluated after winsorising the top and bottom 1 percent of realised and predicted volatility observations. Third, the test sample is divided into low- and high-volatility regimes to evaluate whether model performance is consistent across different market states. All tests for the NN were run for 1,000 epochs, as in the main analysis. For clarity, only the OHLC NN is tested here.

5.1 Removing the Tariff Spike

The first robustness check excludes the April 2025 tariff-related volatility spike from the dataset before the models are estimated. This period is treated as an exceptional exogenous market event because the Trump administration's tariff announcements in early April 2025 triggered a sharp increase in uncertainty and market volatility. Since such an episode can strongly influence model estimation, it is important to test whether the baseline results depend on this short but highly volatile period (Neely, 2025).

The observations from April 2 to April 10 are removed from the training data, after which the models are re-estimated and evaluated on the unchanged test period. This means that the GARCH parameters, as well as the learned weights of the NN model, are estimated without exposure to the tariff-related volatility peak.

Table 9: Robustness Results After Removing the April 2025 Volatility Spike

Model	MAE	RMSE	R^2
NN	0.0285	0.0428	0.5026
GARCH	0.0316	0.0447	0.4867

The results in Table 9 remain broadly consistent after excluding the tariff-related volatility spike. The NN obtains an MAE of 0.0285 and an R^2 of 0.5026, while the GARCH model obtains an MAE of 0.0316 and an R^2 of 0.4867. Compared with the baseline results, performance does not deteriorate substantially. This indicates that the main findings are not solely driven by the April 2025 volatility spike. The NN continues to outperform GARCH slightly, although the difference remains modest.

5.2 Winsorised Evaluation

The second robustness check applies a winsorised evaluation procedure. Volatility data are typically right-skewed and can contain extreme observations during periods of market stress. These extreme observations may disproportionately influence forecast evaluation metrics, especially RMSE, because squared-error metrics penalise large errors more heavily than smaller errors.

To reduce the effect of extreme observations, the top and bottom 1 percent of realised and predicted volatility values are winsorised within each model's test sample. This means that observations above the 99th percentile are set equal to the 99th percentile, and observations below the 1st percentile are set equal to the 1st percentile. The models are then re-evaluated using the same performance metrics as in the baseline analysis.

To ensure that the winsorised evaluation and the tariff-spike removal test are not capturing the same effect, the overlap between both sets of observations is examined. Out of 14,004 total observations, 216 observations are removed in the tariff-spike test and 282 observations are affected by the winsorisation procedure. The overlap between both groups is 88 observations, meaning that 40.7 percent of the removed-peak observations are also winsorised, while only 31.2 percent of the winsorised observations fall within the removed-peak period. This indicates that the two robustness checks are related but not identical and therefore test different sources of sensitivity in the forecasting results.

Table 10: Robustness Results Using Winsorised Evaluation

Model	MAE	RMSE	R^2
NN	0.0327	0.0479	0.3628
GARCH	0.0356	0.0463	0.4906

The winsorised evaluation indicates that the NN is more sensitive to the treatment of extreme observations than the GARCH model. After winsorisation, the NN records an MAE of 0.0327 and an R^2 of 0.3628. Its MAE increases by approximately 0.0052 relative to the original model, while its explanatory power declines substantially. This suggests that part of the NN's baseline performance depends on information contained in the tails of the volatility distribution. In contrast, GARCH achieves an MAE of 0.0356 and an R^2 of 0.4906, remaining broadly consistent with its original performance. Overall, the results suggest that GARCH is more robust to outlier treatment, whereas

the NN's forecasting performance deteriorates when extreme observations are limited.

5.3 High- and Low-Volatility Regime Analysis

The third robustness check evaluates model performance across different volatility regimes. The test sample is divided into low- and high-volatility days based on realised volatility, rather than classifying each individual 5-minute observation separately. This approach is used because volatility regimes are typically market-state conditions that persist over a trading day, while classifying individual 5-minute intervals could create noisy and fragmented regime labels.

This check is important because a model may perform well on average while performing poorly during periods of elevated volatility, when forecasting accuracy is most relevant for risk management. By comparing errors across the two regimes, the analysis assesses whether model performance remains stable across calmer and more volatile market conditions.

Table 11: Robustness Results Across Volatility Regimes

Model	Regime	MAE	RMSE	R^2
NN	High volatility	0.0485	0.0730	-0.2069
NN	Low volatility	0.0210	0.0287	0.0585
GARCH	High volatility	0.0508	0.0654	0.3047
GARCH	Low volatility	0.0223	0.0277	0.2163

The regime analysis shows that model performance differs substantially across market states. In the high-volatility regime, forecast errors are much larger in absolute terms. The NN obtains an MAE of 0.0485, while GARCH obtains an MAE of 0.0508. In the low-volatility regime, the MAE is much smaller, at approximately 0.0210 for the NN and 0.0223 for GARCH. This is realistic because forecasting errors naturally become larger when the level of volatility itself is higher.

The R^2 values are substantially lower in the regime-specific tests than in the full-sample evaluation. For GARCH, the original test R^2 is approximately 0.48, while it decreases to 0.30 in the high-volatility regime and 0.22 in the low-volatility regime. For the NN, the high-volatility R^2 is negative, indicating that the model performs worse than a simple mean forecast within that regime. This does not necessarily imply that the absolute forecasts are unusable, but rather that the model has limited explanatory power once the sample is restricted to a narrower and smaller regime-specific distribution.

5.4 GARCH Parameters

Table 12: Estimated GARCH(1,1) Parameters Across Robustness Tests

Dataset	μ	ω	α_1	β_1
Original	0.0010	0.0003	0.2002	0.7798
Removed peak month	0.0011	0.0002	0.2000	0.7800
Winsorised 1–99	0.0010	0.0003	0.2002	0.7798
High volatility	-0.0016	0.0016	0.4603	0.5397
Low volatility	0.0023	0.0003	0.1246	0.7217

The GARCH parameter estimates confirm that the regime-specific subsets are not simply smaller versions of the original sample, but represent different volatility environments. In the original sample, the GARCH parameters are $\alpha_1 = 0.2002$ and $\beta_1 = 0.7798$, indicating a relatively persistent volatility process. After removing the tariff spike, the estimates remain very similar, with $\alpha_1 = 0.2000$ and $\beta_1 = 0.7800$. This supports the conclusion that the baseline GARCH dynamics are not materially altered by excluding the April 2025 event window.

The high-volatility regime shows a much higher α_1 of 0.4603 and a lower β_1 of 0.5397. This indicates that volatility reacts more strongly to recent shocks and is less persistent than in the full sample. In contrast, the low-volatility regime has a lower α_1 of 0.1246, suggesting weaker sensitivity to new shocks. These parameter shifts help explain why the regime-specific forecasting results are weaker. A single model estimated within a narrower volatility regime has less variation to learn from, fewer observations, and different underlying dynamics.

The standard errors and p -values reported in Appendix E indicate that the GARCH coefficients are statistically significant in nearly all robustness specifications. In the original and winsorised samples, β_1 is highly significant, while α_1 is only significant at the 10 percent level, suggesting that volatility persistence is estimated more precisely than the immediate response to new shocks. After removing the peak month, both α_1 and β_1 become highly significant and their standard errors decline substantially, indicating more precise parameter estimates. The high-volatility regime also produces significant GARCH coefficients, but with larger standard errors, reflecting greater estimation uncertainty. By contrast, the low-volatility regime yields relatively small standard errors and highly significant coefficients, suggesting more stable parameter estimates.

5.5 Overall Robustness

Overall, the robustness checks indicate that GARCH is more stable than the NN across alternative samples and volatility conditions. Removing the April 2025 tariff-related spike has little effect on either model, but the winsorised evaluation shows a clear deterioration in NN performance: its MAE increases and its R^2 falls substantially, whereas the GARCH results remain close to the baseline. The regime-specific tests reinforce this pattern. GARCH retains positive explanatory power in both high- and low-volatility regimes, while the NN records a negative R^2 in the high-volatility regime. Taken together, these findings suggest that the NN achieves competitive baseline performance but is more sensitive to extreme observations and changes in the test distribution, whereas GARCH provides more consistent forecasts across robustness settings.

A potential explanation lies in the large difference in model complexity. The GARCH model contains only four estimated parameters, while the NN contains 121,541 trainable parameters. The parsimonious structure of GARCH directly models volatility clustering and persistence and may therefore generalise better when the sample or market regime changes. In contrast, the NN's much greater flexibility allows it to capture complex nonlinear relationships, but also increases the risk that it becomes dependent on sample-specific patterns and extreme observations. This may explain why the NN performs well in the baseline setting but deteriorates more strongly under winsorisation and regime-specific evaluation.

6 Conclusion

This thesis investigated whether a neural-network framework augmented with high-frequency option-market information can improve short-horizon volatility forecasts relative to traditional GARCH models. Using five-minute S&P 500 index data and features derived from SPX option quotes, the study compared a standard GARCH(1,1) model, a GARCH-X model containing lagged implied volatility, a GRU neural network using only recent OHLC observations, and an augmented GRU neural network combining OHLC observations with engineered return-, calendar-, range-, and option-based features.

The empirical findings show that the neural-network models achieved better forecasting performance than the traditional econometric benchmarks in the main out-of-sample evaluation. However, the incremental improvement obtained by adding option-based and other engineered features to the OHLC neural network was small. The central research question can therefore be answered conditionally: a high-frequency option-data-augmented neural network can improve volatility forecasts relative to traditional GARCH models in the examined sample, but the results do not provide conclusive evidence that the improvement is specifically caused by the option-market variables. A substantial part of the performance gain appears to arise from the neural network's ability to model nonlinear and short-term patterns in the underlying OHLC data.

6.1 Key Findings

6.1.1 Performance of GARCH and GARCH-X

How well do GARCH and GARCH-X forecast volatility?

The standard GARCH(1,1) model provided a meaningful benchmark for five-minute volatility forecasting. Its estimated parameters indicated a highly persistent volatility process, with the sum of the ARCH and GARCH coefficients close to one. This is consistent with volatility clustering, as shocks to volatility were found to decay slowly over time. Out of sample, the GARCH model achieved an MAE of 0.0357, an RMSE of 0.0471, and an R^2 of 0.4833. It therefore substantially outperformed the constant-mean baseline and also improved on the last-value baseline in terms of explanatory power.

Nevertheless, the GARCH forecasts were relatively smooth and did not fully reproduce sharp movements in realised volatility. The model tended to underestimate sudden volatility spikes and to overestimate volatility during particularly calm intervals. This reflects a general limitation of

recursive conditional-variance models: their forecasts are mainly determined by lagged shocks and lagged conditional variance, which causes them to adjust gradually when market conditions change abruptly.

The GARCH-X model added lagged at-the-money implied volatility to the conditional-variance equation. The estimated implied-volatility coefficient was positive and statistically significant, indicating that option-implied volatility contained information related to subsequent conditional variance. The model also produced a slightly better in-sample RMSE and R^2 than the standard GARCH specification.

However, this improvement did not carry over to the test sample. The GARCH-X model recorded an MAE of 0.0412, an RMSE of 0.0520, and an R^2 of 0.4703, performing worse than the standard GARCH model on all three out-of-sample measures. Consequently, the inclusion of implied volatility improved the model's in-sample fit but did not produce more accurate forecasts of unseen observations. The answer to the first sub-question is therefore that GARCH provides a reasonably competitive short-horizon benchmark, whereas the examined GARCH-X specification does not offer a consistent out-of-sample improvement.

6.1.2 Incremental Value of the Augmented Neural Network

To what extent does including option-based features improve NN forecasts relative to using only underlying-asset OHLC and return data?

Both neural-network specifications achieved strong out-of-sample forecasting performance, as shown in Table 13. The addition of the engineered features reduced MAE by approximately 1.5 percent and RMSE by approximately 3.5 percent, while increasing R^2 by approximately 0.034.

To assess whether this difference in predictive accuracy was statistically significant, a Diebold–Mariano test was applied to the out-of-sample squared forecast errors. The test showed that the null hypothesis of equal predictive accuracy was rejected, providing evidence that the augmented neural network had significantly lower squared forecast loss. Nevertheless, the small reductions in MAE and RMSE indicate that the practical magnitude of this improvement remains modest.

Moreover, the augmented feature set contained both option-market variables and other predictors, including lagged returns, calendar indicators, and the previous high-low range. The observed improvement therefore cannot be attributed entirely to option information. More complex option-market features, such as implied-volatility skew, smile curvature, bid-ask spreads, and additional

volume measures, also failed to improve performance consistently. The augmented model thus performs statistically better in the main test sample, but the results provide only limited evidence that option-based features alone produce a substantial or practically meaningful improvement over the information already contained in the OHLC sequence.

Table 13: Out-of-Sample Forecasting Performance

Model	MAE	RMSE	R^2
GARCH	0.0357	0.0471	0.4833
NN OHLC	0.0275	0.0426	0.5074
NN OHLC + Features	0.0271	0.0411	0.5412

6.1.3 Performance of the Neural Network Relative to GARCH

Does a neural network model that incorporates both option data and OHLC data outperform traditional econometric models in forecasting volatility?

The out-of-sample results show that both neural-network specifications outperformed the standard GARCH model in the main test sample. Relative to GARCH, the OHLC NN reduced MAE by approximately 23 percent and RMSE by approximately 10 percent, while producing a moderately higher R^2 . The superiority of the neural network over GARCH is not only visible in MAE, RMSE, and R^2 but is also confirmed statistically. A Diebold–Mariano test rejects equal predictive accuracy ($DM = 6.82$, $p < 0.001$), providing formal evidence that the performance gap is not attributable to sampling variation. The augmented NN achieved reductions of approximately 24 percent in MAE and 13 percent in RMSE relative to GARCH.

These findings suggest that the GRU-based NNs were better able to capture the nonlinear and short-term temporal patterns present in high-frequency volatility data. In contrast, the GARCH model generated smoother forecasts and relied primarily on lagged returns and conditional variance. Although it successfully captured volatility persistence and clustering, it reacted more slowly to abrupt changes and frequently underestimated sharp volatility spikes.

The relative advantage of the NN remained present after removing the April 2025 tariff-related volatility spike. In this robustness test, the NN obtained an MAE of 0.0285 and an R^2 of 0.5026, compared with an MAE of 0.0316 and an R^2 of 0.4867 for GARCH. This indicates that the NN's superior baseline performance was not entirely driven by this exceptional market event.

However, the robustness tests show that the NN does not consistently outperform GARCH across all evaluation procedures. After winsorisation, the NN still achieves a lower MAE, but GARCH performs better in terms of RMSE and R^2 . This indicates that the NN's advantage depends on the selected performance metric and becomes less convincing when extreme observations are limited. In the regime-specific analysis, the NN also retains slightly lower MAE values in both regimes, but GARCH achieves lower RMSE values and higher R^2 values. In particular, the negative R^2 of the NN in the high-volatility regime indicates weak explanatory power within that subsample.

The answer to the third sub-question is therefore that the neural-network models outperform the standard GARCH model in the main out-of-sample evaluation. Nevertheless, this advantage is moderate and not universal across all robustness tests. The results support the use of neural networks for high-frequency volatility forecasting, but they do not establish that neural networks will consistently outperform GARCH across all market regimes and evaluation procedures.

6.1.4 Overall Conclusion

Taken together, the findings show that model flexibility is more important than the mere inclusion of option information. The GRU models outperform the GARCH benchmarks primarily because they can capture nonlinear relationships and short-term temporal patterns in high-frequency data. Option-based information appears to add some predictive value, but this contribution is secondary to the improvement associated with the NN architecture itself.

The study therefore provides qualified support for using machine-learning models in high-frequency volatility forecasting. It does not demonstrate that an option-augmented neural network will always outperform conventional models, nor does it establish that option features are the principal source of the improvement. Rather, it shows that, within the examined SPX sample and forecasting design, a GRU-based framework can provide more accurate forecasts than GARCH, while the addition of option-derived features creates only a modest further gain.

6.2 Limitations

Several limitations should be considered when interpreting the findings.

First, the dataset covers only the period from February to December 2025 and includes a data gap between March 17 and April 4. The effective sample therefore represents less than one full calendar year. Although the dataset contains a large number of five-minute observations, these

observations are not independent and cover a relatively limited number of trading days. The results may consequently reflect the particular market conditions of 2025 rather than stable relationships that generalise across years and economic cycles.

Second, the analysis is restricted to the S&P 500 index and SPX options. SPX options are highly liquid and may contain more reliable information than options on individual stocks or less actively traded indices. The conclusions may therefore not generalise to assets with lower liquidity, wider bid–ask spreads, different trading hours, or different volatility dynamics.

Third, the study uses a single fixed chronological train–test split. This approach prevents look-ahead bias and provides a realistic out-of-sample evaluation, but the reported model ranking may be affected by the particular cut-off date and the market regimes contained in the training and test samples. The robustness tests show that the relative performance of the models changes when extreme observations are treated differently or when the sample is divided into volatility regimes.

Fourth, the option and underlying-index observations are not perfectly synchronised. Option quotes are published with an approximate delay of 22 seconds, and observations from the two datasets are matched to the closest available timestamp. Although the preprocessing procedure corrects the most important timing inconsistencies, residual asynchrony may weaken the measured relationship between option-market information and subsequent realised volatility.

Fifth, the final option representation is relatively parsimonious. It mainly includes at-the-money implied volatility for selected maturities, total volume, and total open interest. This approach does not preserve the complete cross-sectional and temporal structure of the option surface. Potentially relevant information contained in moneyness-specific prices, maturity-specific liquidity, volatility skew, smile dynamics, and changes in the implied-volatility surface may therefore have been lost through aggregation.

Sixth, the GARCH benchmarks are deliberately parsimonious. The standard model assumes Gaussian innovations and does not account explicitly for leverage effects, heavy-tailed innovations, intraday seasonality, structural breaks, or regime changes. The GARCH-X specification also includes only one aggregated implied-volatility variable. The comparison therefore demonstrates an improvement over the selected benchmarks, but not necessarily over every possible econometric volatility model.

Seventh, the neural-network architecture and hyperparameter search are limited by computational considerations. Bayesian optimisation was conducted over a restricted search space, and

neural-network training is affected by stochastic weight initialisation and batch formation. Although validation checkpoints were used, performance from a single fitted network may not fully represent expected performance across repeated training runs.

Finally, although a Diebold–Mariano test was used to compare the two neural-network specifications under squared-error loss, the remaining model comparisons are primarily based on MAE, RMSE, and R^2 . In addition, statistical forecast accuracy does not necessarily imply economic value. The study does not evaluate whether the forecasts improve option trading, portfolio allocation, hedging, or risk-management decisions after accounting for transaction costs.

6.3 Future Research

Future research could extend this study in several directions.

A first extension would be to use a longer sample covering multiple years and a wider range of market conditions. Including calm periods, financial crises, monetary-policy changes, and other structural events would make it possible to examine whether the NN advantage remains stable across economic cycles. A longer dataset would also support rolling-window or expanding-window evaluations with multiple forecast origins rather than relying on one train–test split.

Second, future studies could test the framework on additional assets, including equity indices from other regions, individual stocks, exchange-traded funds, currencies, commodities, and cryptocurrencies. A cross-sectional analysis would help determine whether option information is most useful for highly liquid index markets or whether similar gains can be obtained in less liquid markets.

Third, the contribution of each feature group could be examined through a systematic ablation study. Separate models could be estimated using OHLC data only, return and range features, implied volatility only, option-activity variables only, and the complete combined feature set. Feature-attribution methods such as permutation importance or SHAP values could additionally be used to identify which variables drive individual forecasts and whether their importance changes across volatility regimes.

Fourth, the complete implied-volatility surface could be incorporated more directly. Rather than reducing the option market to a limited number of aggregate variables, future models could use a grid of strikes and maturities, surface parameters, or latent factors extracted through principal-component analysis or an autoencoder. Architectures such as convolutional networks, attention mechanisms, or transformers may be better suited to learning the cross-sectional structure of the option surface.

Fifth, alternative forecast targets and horizons should be considered. The present study focuses on one-step-ahead five-minute volatility. Option-market information may be more valuable over longer horizons because implied volatility reflects expectations over the remaining life of an option rather than only the next five-minute interval. Comparing five-minute, hourly, daily, weekly, and maturity-matched forecasts could reveal the horizons at which option data provide the greatest incremental value.

Sixth, future research should use repeated model estimation and forecast-comparison tests. NN results could be averaged over multiple random seeds, while additional Diebold–Mariano tests or model-confidence-set procedures could be used to assess whether differences in forecast accuracy are statistically significant. Evaluation across multiple rolling test windows would further reduce dependence on one particular market period.

Finally, future studies should examine the economic usefulness of the forecasts. Forecasts could be incorporated into option-pricing, delta-hedging, volatility-trading, portfolio-allocation, or risk-management strategies. Comparing realised profits, hedging errors, value-at-risk exceedances, and transaction-cost-adjusted performance would establish whether the statistical improvements documented in this thesis translate into economically meaningful benefits.

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Appendix A Sustainable Development Goals (SDG) Statement

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Date: August 31, 2026

Through the research conducted for this master's thesis, I seek to contribute to one or more of the 17 SDGs set forth by the United Nations: <https://www.undp.org/sustainable-development-goals>.



SDG Code(s): 8

Explanation: This thesis links to SDG 8 because better volatility estimation can improve risk management and market stability. More accurate option-based volatility forecasts can reduce hedging costs and prevent inefficient capital allocation during stress, supporting sustainable economic activity. The collaboration with Tickstream also contributes to innovation and high-quality fintech development.

Appendix B Additional Data Exploration

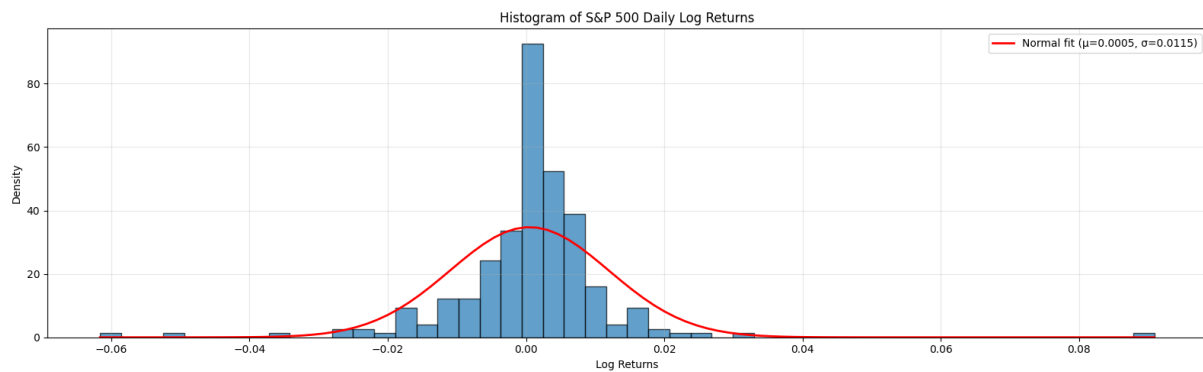


Figure 7: Distribution of daily S&P 500 log returns. The histogram is compared with a fitted normal distribution, illustrating the concentration of returns around zero and the presence of extreme observations in the tails.

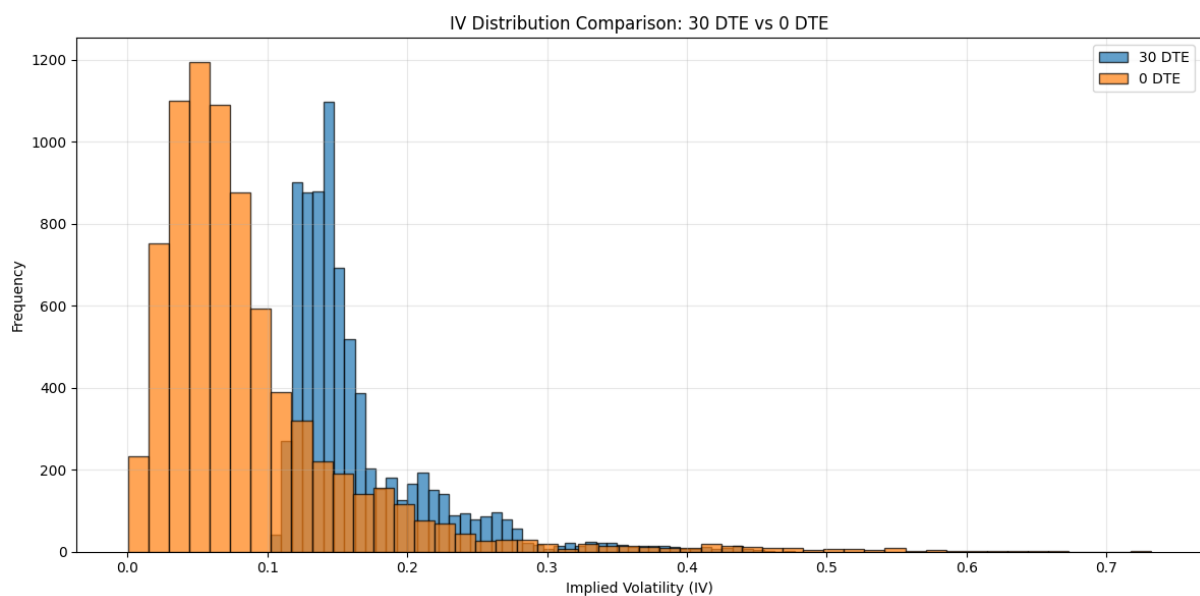


Figure 8: Distribution of at-the-money implied volatility for options with approximately 30 days and 0 days to expiration. The distributions differ in level and dispersion, with 0-DTE implied volatility exhibiting greater concentration at lower values and a longer right tail.

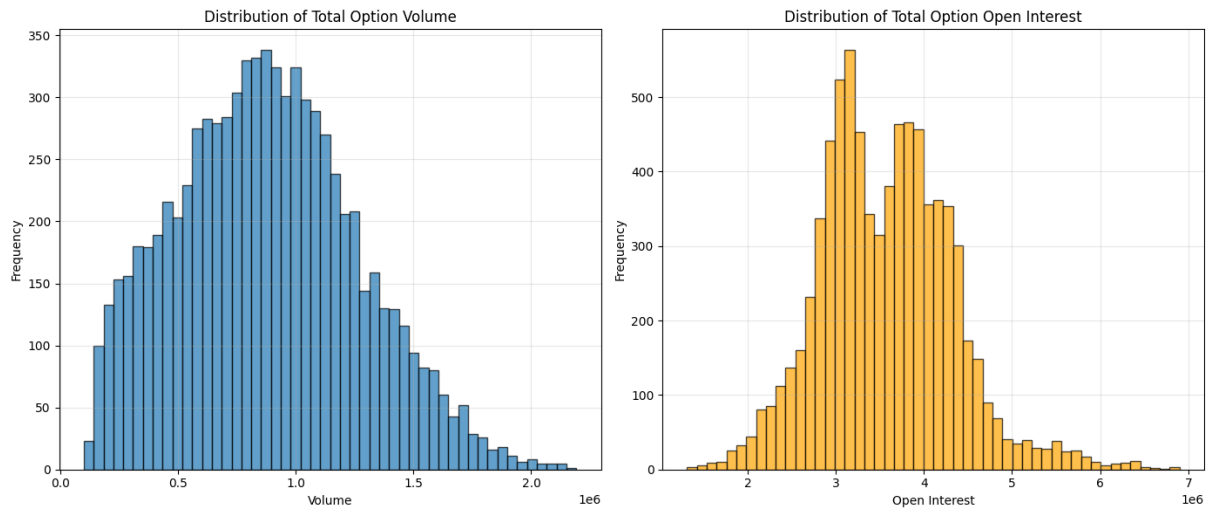


Figure 9: Distribution of total option trading volume and total open interest across the sample.

Appendix C Error measures GARCH

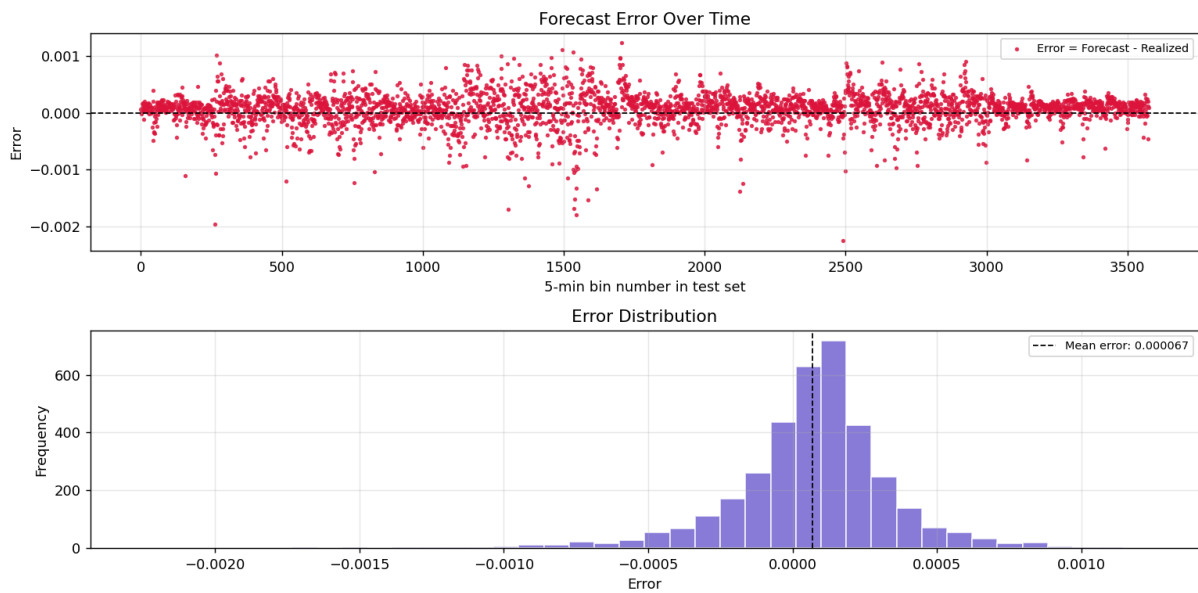


Figure 10: Error measures GARCH

Figure 10 presents both the time series of forecast errors and their distribution for the GARCH(1,1) model. The upper panel shows the evolution of forecast errors over time. The errors fluctuate around zero, indicating that the model is approximately unbiased. The lower panel displays the distribution of forecast errors. The distribution is centered close to zero, further confirming the absence of systematic bias. However, it exhibits fat tails and slight asymmetry, indicating that large forecast errors occur more frequently than would be expected under a normal distribution. This aligns with the heavy-tailed nature of financial volatility and suggests that extreme events are not fully captured by the model.

Appendix D Forecast Figure

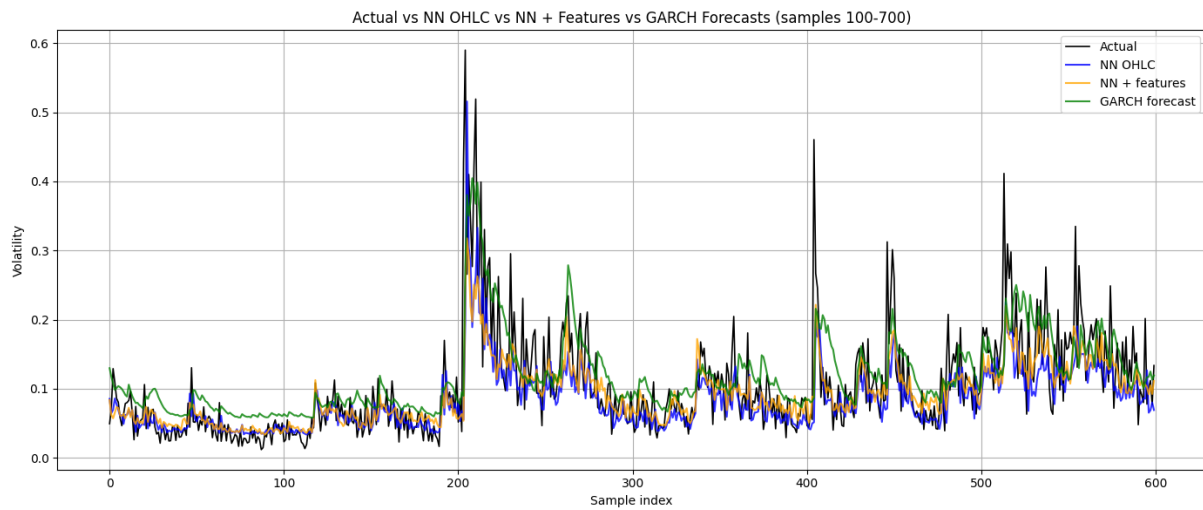


Figure 11: Actual vs NN OHLC vs NN + Features vs GARCH Forecasts

Figure 11 compares realised volatility with the forecasts of the GARCH model, the OHLC-only NN, and the augmented NN. All models capture the general clustering of volatility, but the NN forecasts follow short-term changes more closely than GARCH. The augmented model appears to track several volatility peaks slightly better, although the differences between the two neural-network specifications are visually modest. Sharp volatility spikes remain difficult for all models to predict accurately.

Appendix E GARCH Robustness Point Estimates

Table 14: GARCH(1,1) parameter estimates across robustness samples

Dataset	μ	ω	α_1	β_1
Original	0.000982	0.000322	0.200191*	0.779814***
Standard error	(0.001100)	(0.000456)	(0.115030)	(0.147651)
<i>p</i> -value	[0.3721]	[0.4810]	[0.0818]	[< 0.0001]
Peak month removed	0.001079*	0.000160***	0.199999***	0.780007***
Standard error	(0.000653)	(0.000055)	(0.022999)	(0.022241)
<i>p</i> -value	[0.0985]	[0.0034]	[< 0.0001]	[< 0.0001]
Winsorised (1st–99th percentile)	0.000982	0.000322	0.200191*	0.779814***
Standard error	(0.001100)	(0.000456)	(0.115030)	(0.147651)
<i>p</i> -value	[0.3721]	[0.4810]	[0.0818]	[< 0.0001]
High-volatility regime	-0.001630	0.001600*	0.460252**	0.539748***
Standard error	(0.002883)	(0.000876)	(0.181457)	(0.143161)
<i>p</i> -value	[0.5718]	[0.0678]	[0.0112]	[0.0002]
Low-volatility regime	0.002270***	0.000327***	0.124582***	0.721716***
Standard error	(0.000566)	(0.000118)	(0.027143)	(0.034241)
<i>p</i> -value	[< 0.0001]	[0.0055]	[< 0.0001]	[< 0.0001]

Notes: Parameter estimates are reported in the first row for each sample, standard errors are shown in parentheses, and exact *p*-values are shown in square brackets. The conditional variance equation is $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Values smaller than 0.0001 are reported as $p < 0.0001$.

Appendix F Official statement of originality and use of AI

By this statement, I hereby acknowledge the submitted thesis, entitled:

High-Frequency Volatility Forecasting with an Option-Data-Augmented Neural Network

to be produced independently by me, without external help. Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is given. By this statement, I explicitly declare that I am aware of the fraud sanctions as stated in the Education and Examination Regulations (EERs) of the SBE.

During the preparation of this thesis, I made use of generative artificial intelligence tools throughout the research and writing process. ChatGPT was used to assist with literature review, grammar, academic phrasing of text and supporting the organisation of content. GitHub Copilot was used within Visual Studio Code to support the coding process, for example by suggesting code, identifying possible errors, and assisting with debugging. All research decisions, analyses, interpretations, and final written content were reviewed and approved by me, and I remain fully responsible for the accuracy and originality of this thesis.

Frans Staats